

● MEDIUM

● GEOMETRY AND TRIGONOMETRY

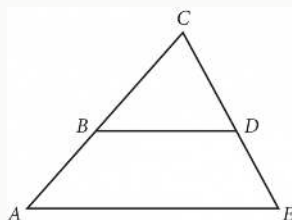
Geometry and Trigonometry

01

10 questions · ~15 min

Q1

GEOMETRY AND TRIGONOMETRY



Note: Figure not drawn to scale.

In the figure above, segments AE and BD are parallel. If angle BDC measures 58° and angle ACE measures 62° , what is the measure of angle CAE ?

- A. 58°
- B. 60°
- C. 62°
- D. 120°

Square X has a side length of 12 centimeters. The perimeter of square Y is 2 times the perimeter of square X. What is the length, in centimeters, of one side of square Y?

- A. 6
- B. 10
- C. 14
- D. 24

Q3

GRID-IN

GEOMETRY AND TRIGONOMETRY

What is the radius of the circle in the xy -plane defined by $x^2 + y^2 = 169$?

STUDENT-PRODUCED RESPONSE

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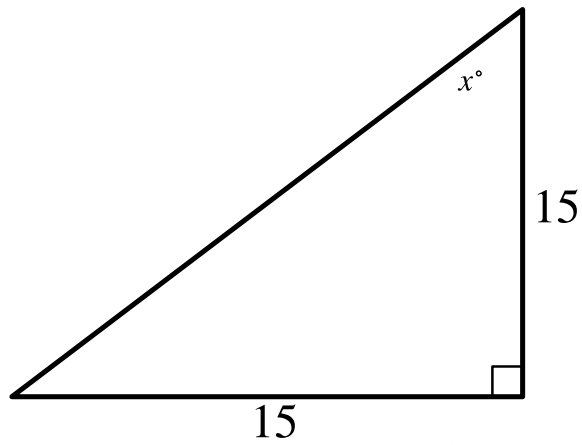
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q4

GRID-IN

GEOMETRY AND TRIGONOMETRY



Note: Figure not drawn to scale.

- One angle is a right angle.
- The measure of a second angle is x° .
- The lengths of the 2 sides adjacent to the right angle are as follows:
 - 15
 - 15
- Note: Figure not drawn to scale.

In the triangle shown, what is the value of x ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q5

GEOMETRY AND TRIGONOMETRY

In triangle ABC, the measure of angle A is 50° . If triangle ABC is isosceles, which of the following is NOT a possible measure of angle B ?

- A. 50°
- B. 65°
- C. 80°
- D. 100°

Q6

GEOMETRY AND TRIGONOMETRY

The length of one side of square M is 6 times the length of one side of square N. The area of square N is 49 square feet. What is the area, in square feet, of square M?

- A. 1,764
- B. 504
- C. 294
- D. 252

The measure of angle Z is 60° . What is the measure, in radians, of angle Z ?

A. $\frac{1}{6}\pi$

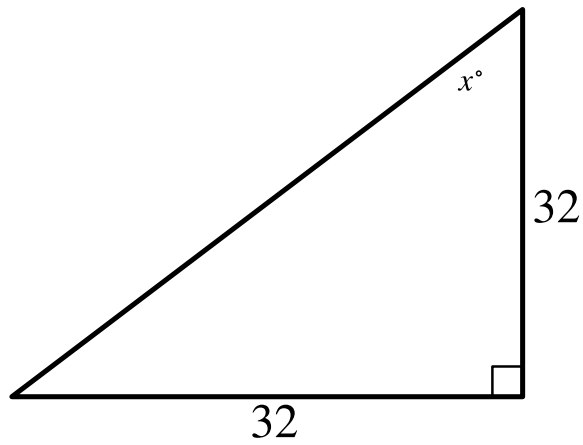
B. $\frac{1}{3}\pi$

C. $\frac{2}{3}\pi$

D. 1π

Q8

GRID-IN GEOMETRY AND TRIGONOMETRY



Note: Figure not drawn to scale.

- One angle is a right angle.
- The measure of a second angle is x° .
- The lengths of the 2 sides adjacent to the right angle are as follows:
 - 32
 - 32
- Note: Figure not drawn to scale.

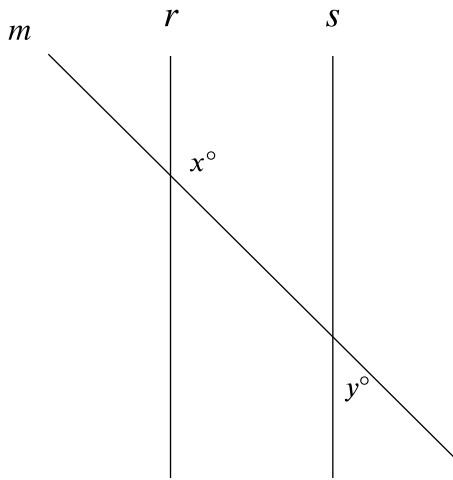
In the triangle shown, what is the value of x ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8



Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.



Note: Figure not drawn to scale.

- Clockwise from top left, the 3 lines are labeled m , r , and s .
- Line m intersects both line r and line s .
- At the intersection of line m and line r , 1 angle is labeled clockwise from top left as follows:
 - Top right: x°
- At the intersection of line m and line s , 1 angle is labeled clockwise from top left as follows:
 - Bottom right: y°
- A note indicates the figure is not drawn to scale.

In the figure shown, lines r and s are parallel, and line m intersects both lines. If $y < 65$, which of the following must be true?

- A. $x < 115$
- B. $x > 115$
- C. $x + y < 180$

D. $x + y > 180$

Q10

GEOMETRY AND TRIGONOMETRY

A triangle has a base length of 10 centimeters and a corresponding height of 70 centimeters. What is the area, in square centimeters, of the triangle?

- A. 700
- B. 350
- C. 175
- D. 80